

BOOTSTRAPPING TTS FOR MASHI:
A FIRST SPEECH DATASET AND XTTS-v2 FINE-TUNING FOR A
ZERO-RESOURCE BANTU LANGUAGE OF DR CONGO

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github.com/Ashuza11/mashi-tts-bootstrap **Poster ID: TODO: insert assigned unique ID**

1. Motivation: 3 million speakers, zero tools

Mashi (Shi, ISO 639-3: **shi**) is a Bantu language spoken by ~3 million people in South Kivu, eastern DRC (Ngweshe, Nyangezi, Bunyakiri, Muganzo). Before this project it had **no** TTS, ASR, NLP dataset, or language model of any kind. Related languages (Kinyarwanda, Kirundi) are themselves low-resource and absent from mainstream TTS models.

Research question

Can fine-tuning a multilingual TTS model (Coqui XTTS v2) on ~**1 hour** of curated Mashi audio produce intelligible Mashi speech?

Speech-first access matters here: much of the speaker population reads French or Swahili rather than written Mashi, so *voice* interfaces are the natural entry point for local-language technology.

2. Contribution 1 — the first Mashi speech dataset

723 clips / **67.4 minutes** of segmented, transcribed, TTS-ready audio (22050 Hz mono WAV, 2.0–11.3 s, cut only at natural pauses), built from ~10 h of raw audio across **3 speakers** and **5 domains**:

Speaker	Domain	Clips	Min
bib (M, ~35)	Bible — Genesis ch. 1–4	183	24.2
dav (M, 51)	Short stories / folk tales	109	12.5
dav	Numbers (esabu)	39	3.5
dav	Family vocabulary	25	3.0
dav	Calendar (nsaha)	31	3.3
mrl (F, ~28)	Clock-time expressions	336	20.9
Total		723	67.4

Split: **687 train** / **36 eval**, stratified per speaker+domain (seed 42). Sources: Padri P. Matabaro’s Mashi Bible audio (bible.com), original texts by the collector read by D. Namulabarha, and murhula.com talking-clock components (permission: M. Nshombo). *Disclosure*: the clock-time domain recombines 95 unique component recordings, the same way the source site plays them.

3. Contribution 2 — a low-cost curation pipeline

- A reproducible recipe for collectors with no lab and no GPU:
- **Silence-based segmentation at natural pauses** — narrative audio (Bible, stories) is split only at detected silences, so no word is ever cut; vocabulary recordings (one item per pause) segment themselves.
 - **Component-based synthesis of clock-time clips** — hour + minute announcements from a public talking clock are assembled into full time expressions, multiplying 10 min of source audio into 336 aligned clips.
 - **Text normalization** — casing and apostrophe policy for a Latin-based orthography with diacritics (ê, â, î, û).
 - **Everything transcribed and verified by a native/heritage speaker**; per-clip provenance and permissions recorded in a data card.

Total cash cost of the dataset: ≈ **\$0** (phone recordings + public sources + consent), plus curation time.

4. Model & training setup

Base: Coqui **XTTS v2** (multilingual, voice-cloning GPT + HiFi-GAN decoder). None of its 17 languages is Bantu, so we use **Spanish as a phonetic proxy**: it shares Mashi’s five pure vowels /a e i o u/ and open CV syllable structure, and its text frontend expands clock digits (“9:05”) deterministically at train *and* inference time.

- Fine-tune the GPT component on 687 clips; eval on 36 held-out clips.
- **10 epochs**, batch $8 \times$ grad-accum 2 (effective 16), max clip 11.6 s.
- Hardware: one **A100** on Modal; total fine-tune wall-clock **< 2 h[†]** — reproducible for ≈ **\$5–10[†]** of cloud credit.
- Voice conditioning at inference from one reference WAV per speaker.

5. Results [†]= estimate from published comparables — final numbers replace these after the training run

Training

TODO: insert train/eval loss curve figure (results/loss_curve.png)
Eval loss expected to fall steadily and plateau by epoch ~8[†]; with 67 min of data the risk is overfitting, monitored via the held-out eval split.

Native-speaker evaluation (no Mashi ASR exists, so automatic CER is impossible — evaluation is human)

Metric (5-pt MOS / %)	In-domain	Out-of-domain
Naturalness MOS	2.5–3.5 [†]	2.0–3.0 [†]
Intelligibility (words correctly transcribed by native listeners)	60–80 % [†]	40–65 % [†]
Speaker similarity MOS	3.0–4.0 [†]	3.0–4.0 [†]

Raters: **TODO: N** native Mashi speakers, **TODO: K** synthesized utterances per domain, 5-point Likert scale.

Why these estimates are plausible

System	Data	Setting	Result
BibleTTS (VITS)	33–87 h	6 African langs.	MOS 2.8–4.3 in-dom. 2.3–3.9 out-dom.
BnTTS (XTTS)	3 850 h + 20 min/spk	Bangla, few-shot speaker adapt.	SMOS 4.62, CER 0.034
Align2Speak	0.5–5 h	unseen langs. via fine-tune + GRPO	CER 33% → 3.9% (Portuguese, 30 min)
Ours	1.1 h	unseen Bantu lang., Spanish proxy	MOS 2.5–3.5[†]

With 30× less data than BibleTTS’s smallest language and no large-scale target-language pre-training (unlike BnTTS), we expect results *below* those systems — the claim under test is **intelligibility**, not studio quality.

Listen

TODO: QR code linking to audio samples (per-speaker: Bible verse, story s...)
TODO: 2–3 mel-spectrograms: ground truth vs. synthesized

6. Limitations & ethics

- **Domain skew**: 36% Bible narrative, 29% templatic clock times; conversational Mashi is underrepresented.
- **Spanish proxy** cannot model Mashi tone or vowel length; prosody errors are expected and reported, not hidden.
- **Consent & attribution**: all three sources recorded with permission or from credited public religious media; the dataset is distributed gated (request-based), not scraped-and-dumped.
- Voice cloning risk mitigated by gated release of speaker reference audio.

7. What’s next

- Fill the coverage gaps (Genesis ch. 13 transcript, ch. 20/25 audio) → ~10 h of aligned Bible speech for **Mashi ASR**, which then unlocks automatic CER for TTS.
- True **cross-lingual transfer from Kinyarwanda**: continue pre-training on Common Voice Kinyarwanda (2000+ h) before Mashi fine-tuning, replacing the Spanish proxy.
- More speakers (target: balanced gender, 3 dialects), conversational domain, and a public gated release on Hugging Face.

8. References

[1] Casanova et al., *XTTS: a Massively Multilingual Zero-Shot TTS Model*, Interspeech 2024. [2] Meyer et al., *BibleTTS: a Large, High-Fidelity, Multilingual and Uniquely African Speech Corpus*, Interspeech 2022. [3] Rakib et al., *BnTTS: Few-Shot Speaker Adaptation in Low-Resource Setting*, NAACL Findings 2025. [4] *Align2Speak: Improving TTS for Low-Resource Languages via ASR-Guided Online Preference Optimization*, arXiv:2509.21718. Audio sources: Padri Pierre Matabaro OFM (bible.com/bible/3953/GEN.1.MKB); murhula.com — Marius Nshombo; recordings by David Namulabarha.